

Project Name: Appointment Scheduling Tracker

Group Number: Group 13

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Team Members: Asif Khan, Genesis Enemuoh, Nensa Prasai

Sponsor: None

New requirements:

NR1: The system will prevent people from booking appointment slots.

NR2: The system will send a confirmation email when an appointment is successfully scheduled.

NR3: The system will allow users to reschedule or cancel an appointment.

NR4: The system will send reminder notifications 24 hours before an appointment.

Change requirements:

CR1: The system shall have a maximum downtime of 6 hours per month.

CR2: The system shall allow appointment reminders to be sent 24 hours before the appointment.

CR3: The system shall support cancellation and rescheduling of appointments.

Updated Schedule:

- Requirements Analysis – Completed (Week 1)
- System Design – Completed (Week 2)
- Prototype Development – In Progress (Week 3)
- Testing – Planned (Week 4)
- Final Documentation – Planned (Week 5)

Updated Budget:

Item	Cost per year
Figma	\$0
UI/UX	\$0
Git Hub	\$0
Email notification reminder	\$35
Front-end design	\$700
Back-end design	\$900

System Architecture Diagram

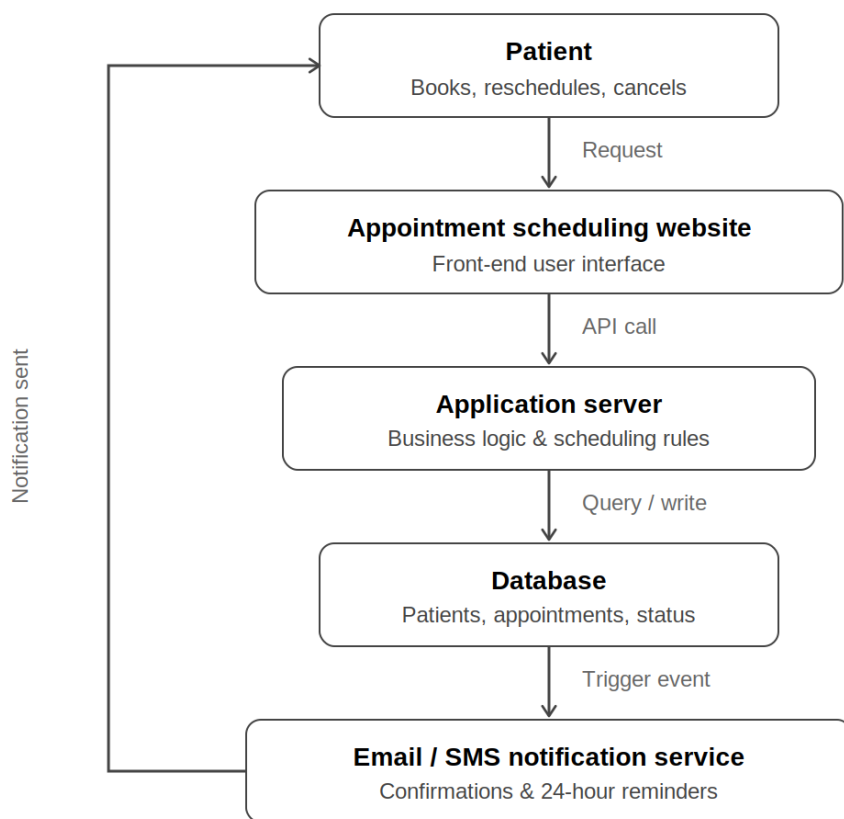


Figure 1: Appointment Scheduling Tracker – System Architecture Diagram

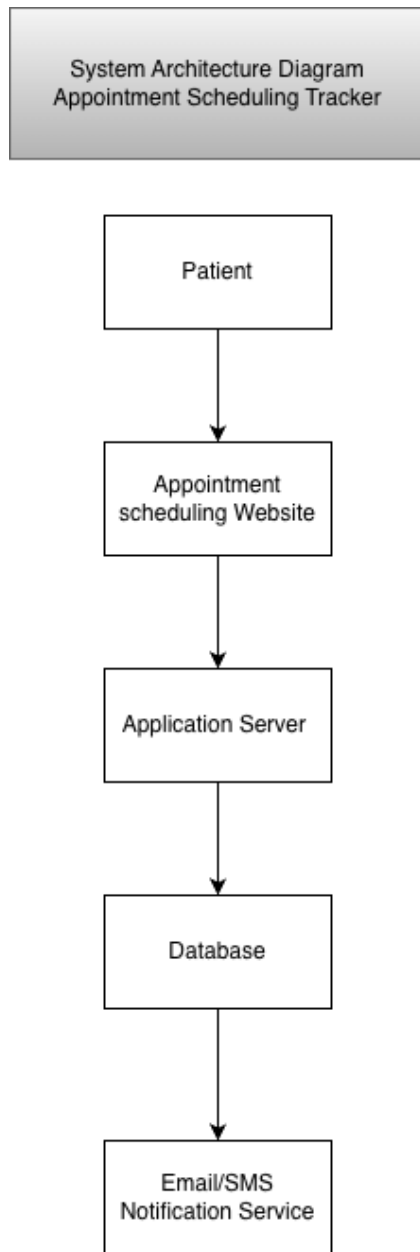


Figure 1 shows the Appointment Scheduling Tracker's system architecture diagram. The graphic illustrates how a patient interacts with the website that schedules appointments and connects with the database and application server. The email/SMS notification service for changes and appointment reminders is supported by the database.

Data Flow Diagram (DFD)

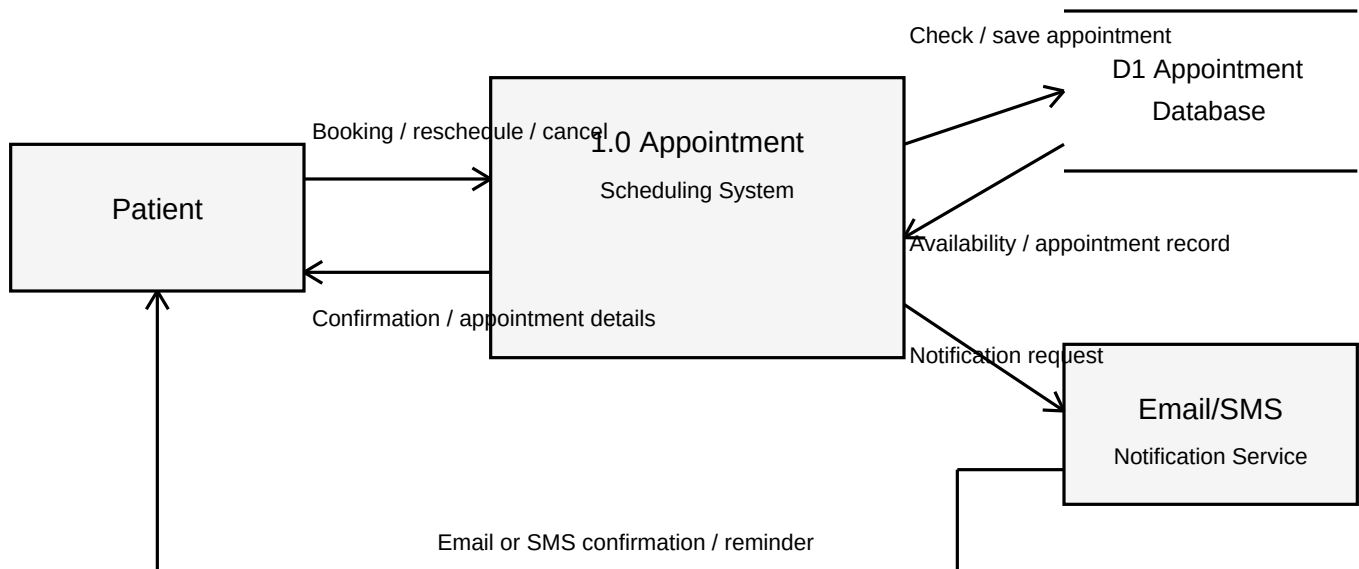


Figure 2 illustrates how appointment data moves through the system. The patient submits a booking, rescheduling, or cancellation request to the Appointment Scheduling System. The system checks and updates the Appointment Database, then sends confirmation and reminder requests to the Email/SMS Notification Service. The patient receives the resulting appointment details and notifications.

Figure 2: Appointment Scheduling Tracker - Data Flow Diagram

Entity Relationship Diagram (ERD)

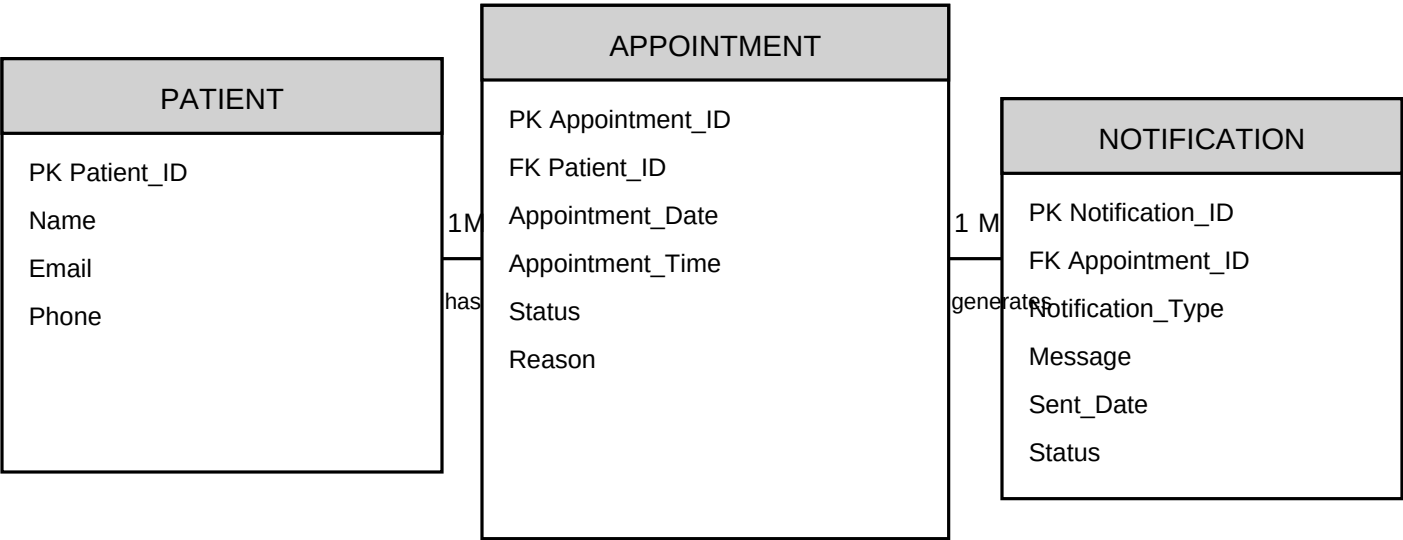


Figure 3 shows the main database entities and their relationships. One patient can have many appointments, while each appointment belongs to one patient. One appointment can generate multiple notifications, and each notification is associated with one appointment. Primary keys uniquely identify records, while foreign keys connect the related entities.

Figure 3: Appointment Scheduling Tracker - Entity Relationship Diagram

DFD and ERD Diagram Descriptions

Data Flow Diagram

The DFD represents the movement of information among the patient, appointment scheduling system, appointment database, and notification service. It shows how requests are received, validated, stored, and returned to the patient as appointment confirmations or reminders.

Entity Relationship Diagram

The ERD defines the database structure for the appointment scheduling system. The Patient entity stores user information. The Appointment entity stores scheduling details and references the patient through Patient_ID. The Notification entity stores email or SMS records and references the related appointment through Appointment_ID.

Relationship Summary

- One Patient can have many Appointments.
- Each Appointment belongs to one Patient.
- One Appointment can generate many Notifications.
- Each Notification belongs to one Appointment.